1

Trocknung

Burkhard Pick

31.10.2026

First Author:

Available at https://github.com/pmichaillat/latex-paper

1. Introduction

Research question. xx

1.1. Generic subsection

Lorem ipsum dolor sit amet,

1.2. Subsection with URL

It is possible to insert URLs, such as the URL for this paper template's repository: `https://github.com/pmichaillat/latex-paper`. Lorem ipsum dolor sit amet, consectetur

2. Section with math

This section displays a number of mathematical expressions to showcase the math fonts used in the template.

2.1. Roman letters

The Roman characters in math are just the same as the characters in the text—but in *italic*. Here are some small letters:

$$a\{p-r\times l\}+\frac{w(g/z+j)}{i(t)+j(t)+k(t)-e^{p}-x^{j}}+\frac{h[f]+x^{f}}{k[y]-e^{y}+x^{y}}=f(j)^{6+y}-i_{g,j,p}^{3}\approx(p_{ji})^{5}.$$

Here are some capital letters as well:

$$G[p+P^{7}-Q]-A_{B}+L\{j\}=F(X)\to[Y+K]\times Z_{f}/H-[g_{4}-i],\text{ for any }i.$$

The punctuation is also the same in math as in text.

2.2. Digits

The digits in math are also taken from the text font, just like the Roman characters. So 1, 2, 3, 4, 89, 03 (in text mode) is just the same as 1, 2, 3, 4, 89, 03 (in math mode). The percentage sign is also the same so 27% (in text mode) is the same as 27% (in math mode).

2.3. Greek letters

2.4. Calligraphic letters

Calligraphic letters are taken from the same family of fonts as the Greek letters. The calligraphic letters are only available for capital letters. Here are some of the calligraphic letters:

$$\frac{\mathcal{A}(A)}{\mathcal{H}} - \mathcal{B}(\beta) + \mathcal{Q}(j + 3f) \to \mathcal{Y} + \mathcal{M}(\mathcal{N} - 2) = \mathcal{J}(X) \neq \mathcal{X}(J) - \mathcal{F}[\mathcal{D} \times \Delta]^{8}.$$

2.5. Blackboard-bold letters

Here are some blackboard-bold letters, used in some mathematical expressions: $x \in \mathbb{R}^n$ but $y \in \mathbb{Q}$ and $[A_1, A_2] \in \mathbb{Z} \times \mathbb{N}$; so that $\mathbb{E}[X \mid Y] = Y + 3 \cdot \mathbb{P}(X^2) \notin \mathbb{C} \times \mathbb{1}(X \subseteq \mathbb{B})$.

2.6. Symbols

The template comes with a full set of mathematical symbols. Here is a random assortment of some symbols:

$$\left(\frac{\sum_i \vec{z}_i}{\sum_{j=-\infty}^{+\infty} X_j} \right) \pm Q \subset \mathbb{R} \neq \mathbb{C} \implies \sqrt{\frac{\chi}{2}} > \sqrt{\mathcal{Y}} \gg \hat{b}(y_0).$$

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed aliquam magna vel urna ultrices, sed lacinia nulla mattis. Fusce non nunc nec est mollis malesuada. Nullam dignissim nulla sit amet libero facilisis, eget fringilla libero sagittis. Suspendisse potenti. Vivamus fermentum consectetur ante, at rhoncus nisi tristique vel:

$$\sum_i \left[\frac{\hat{x}(i)}{\mathcal{Y}(i)} \right]^2 \leq 2\pi \Rightarrow \prod_j \bar{y}(j)^2 \geq \left\{ \iint \frac{g(h)}{\gamma(s) \times \Delta(h)} \, dh \, ds \right\} \mapsto f \circ g(x) \iff \tilde{Q}(x) \propto \frac{\dot{x}}{2}.$$

2.7. Bold characters

In the template it is possible to bold all math characters. Roman characters can be bolded, such as $\boldsymbol{a} + \boldsymbol{D} = \boldsymbol{E}^2 + \boldsymbol{j}/\boldsymbol{i}$. Greek characters can also be bolded, such as $\boldsymbol{\alpha} + \boldsymbol{\Delta} = \boldsymbol{\epsilon}^2 + \boldsymbol{\Lambda}/\boldsymbol{\Phi}$. It is also possible to bold digits: $1 + 2 \neq \mathbf{1} + \mathbf{2}$. Finally, it is possible to bold calligraphic letters: $\boldsymbol{\mathcal{C}} + \boldsymbol{\mathcal{E}} - [\boldsymbol{\mathcal{X}} + \boldsymbol{\mathcal{Y}}]$. Blackboard-bold letters are already “bold” so they cannot be bolded further.

2.8. Operators

Operators in mathematical expressions are typeset with the text font. Here are a few examples:

$$\max_{x} \pi(x) = \frac{\int \ln(u) du}{\int \exp(v)\, dv} \mapsto \cos(\theta) + \sin(\omega) - \min_{z}[\tan(z)].$$

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Aliquam erat volutpat. Donec tincidunt, quam sed pellentesque fermentum, lorem dolor consequat lectus, at pulvinar arcu ipsum in ligula. Mauris pretium ipsum id sapien posuere vehicula.

2.9. Some equations

This subsection illustrates how the different math fonts fit together. The equations below combine various characters together.

First, here are equations in text. Lorem ipsum dolor sit amet, consectetur adipiscing elit: $\prod_{i} \alpha_{i}(X) \geq 2 \cdot [\partial a_{u}(Z)/\partial Z] \propto \mathcal{X} \sim 5.2\pi \rightsquigarrow +\infty$. Lorem ipsum dolor sit amet, consectetur adipiscing elit: $\sum_{x=1}^{4} \xi^{x}(j) = 0, \ \forall j \in \mathbb{R}$. Here are some other expressions: $\mathcal{R}(z) = \max\{T_1, \ldots, T_n\} > \hat{z}$, and $\mathbb{P}(T_n < z) = 1 - S(z) = F(z)$. Sed vehicula ultrices odio, vitae maximus justo convallis at. Fusce quis consequat arcu: $\alpha^{*} = S(z^{*})$ and $S^{*}(z^{*}) = \mathbb{E}[\boldsymbol{\alpha} \mid \beta]$. Integer non ante eget purus aliquet consequat. Mauris ut dui nec ligula consectetur commodo.

Here are equations in display mode, without labels:

$$\frac{\mathcal{S}(u^{*})}{1 - \frac{\partial \ln(\theta)}{\partial \theta} + [\gamma^{\eta}] - \mathcal{G}(z^{*})} = \sum_{i=0}^{+\infty} \frac{dF^{i}(x)}{dx} - \Lambda(i) \geq \left\{ \sum_{i=0}^{+\infty} G^{i} - \Omega(i) \right\}$$
$$\frac{\mathbb{Q}(z^{*})}{1 - \mu + \exp(\xi) \times \mathbb{M}(z^{*})} = \prod_{i=0}^{+\infty} F^{i} - \Phi(i) \neq \left(\int_{0}^{+\infty} \frac{G^{i}}{\epsilon(i)} di \right).$$

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed vehicula ultrices odio, vitae maximus justo convallis at. Fusce quis consequat arcu. Integer non ante eget purus aliquet consequat. Mauris ut dui nec ligula consectetur commodo.

And here is a single equation in display mode without a label:

$$\dot{y}(t) \to \sum_{j=0}^{+\infty} \mathcal{W}^{j}(z^{*}(t)) = \frac{S(z)\mathbb{E}(N(z))}{\mu_1 - \zeta_2} \cdot 2 - \left[\prod_{i=0}^{+\infty} \mathcal{F}^{i} \right] - \iiint \exp(\lambda(t)\mu(s)) \sin(\theta) dt\, ds\, d\theta.$$

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac

turpis egestas. Nam consequat, ipsum eget tincidunt aliquam, mi est pellentesque lacus, nec ullamcorper leo magna a dui.

2.10. Theorems, propositions, lemmas, and other results

Here is a proposition with some more math:

PROPOSITION 1. *Lorem ipsum dolor sit amet, consectetur adipiscing elit:*

$$\sum_{k} \boldsymbol{S}_{k_x}(z) \approx \frac{S(z)^x}{k/23 - \zeta\gamma[45 - S(z)] + \ln(y) - j^2 + x(l)}. \tag{1}$$

Ut enim ad minim veniam, sunt in culpa qui officia deserunt mollit anim id est laborum. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

PROOF. Here is the proof to the proposition. Donec commodo justo a eros malesuada, eget vulputate tortor accumsan. Sed ac pulvinar nulla. Etiam quis felis dapibus, vulputate metus eu, finibus nunc. Sed vel sodales dui. Nam venenatis dolor non orci tempus fermentum. Vivamus sodales justo a ligula cursus aliquet. Sed fringilla nunc vitae justo finibus, id placerat lectus sodales. □

Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia curae; Sed ac eros vel felis vehicula vehicula. Phasellus interdum justo a felis congue, vel dapibus est tristique. Now here is a lemma—note that all types of results have their own counter:

LEMMA 1. *Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat:*

$$z^* = \int_0^{\infty} \alpha(i) \cdot \frac{1 - \beta}{1 - \alpha(i)\beta} \, di. \tag{2}$$

Quia dolor sit amet consectetur adipiscing velit, sed quia non numquam eius modi tempora incidunt, ut labore et dolore magnam aliquam quaerat voluptatem. Then here is another lemma:

LEMMA 2. *Consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua* $\mathcal{Z}(\alpha)$.

$$\frac{\sum_i z^i}{\prod_i q^i} \to \frac{\int_0^{\infty} \alpha(i) \cdot [1 - \beta] \, di}{1 - \exp(\alpha)\sin(\beta)}.$$

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

Pellentesque

Nemo enim ipsam voluptatem, quia voluptas sit, aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos, qui ratione voluptatem sequi nesciunt, neque porro quisquam est, qui dolorem ipsum. And here is a theorem following the lemma:

THEOREM 1. *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua:*

$$\mathbb{E}(N(z^*)) \approx \frac{1 - \mathbb{P}(\alpha\pi)}{1 - \pi} - \frac{f(y)}{z(p)^*} + P(\Gamma).$$

Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Nemo enim ipsam voluptatem, quia voluptas sit, aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos, qui ratione voluptatem sequi nesciunt, neque porro quisquam est, qui dolorem ipsum. And here is a corollary following the theorem:

COROLLARY 1. *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua:*

$$\mathbb{Q}^* = \int_{-\infty}^{+\infty} \mu(j)^2 \cdot \frac{1 - \ln(\nu(j))}{1 - \exp(\beta(j))}\, dj.$$

Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

3. Section with lists

Lists are not very common in scientific papers, but they are sometimes useful. This section presents a few possible lists.

3.1. Subsection with numbered lists

And here is a numbered lists with two levels:

1. Et harum quidem rerum facilis est et expedita distinctio.
2. Nam libero tempore, cum soluta nobis est eligendi optio.

3. Emporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet. Nam libero tempore, cum soluta nobis est eligendi optio:
    A. Cumque nihil impedit.
    B. Quo minus id, quod maxime placeat.
    C. Facere possimus, omnis voluptas assumenda est.
4. Et harum quidem rerum facilis est et expedita distinctio.
5. Nam libero tempore, cum soluta nobis est eligendi optio.

Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat.

4. Section with graphs

Here is a section with a variety of graphs. At vero eos et accusamus et iusto odio dignissimos ducimus, qui blanditiis praesentium voluptatum deleniti atque corrupti, quos dolores et quas molestias excepturi sint, obcaecati cupiditate non provident, similique sunt in culpa, qui officia deserunt mollitia animi, id est laborum et dolorum fuga.

4.1. Subsection with graphs at the top of the page

A simple two-panel graph is on figure **??**. It will be placed at the top of the page, just about here. Et harum quidem rerum facilis est et expedita distinctio. Nam libero tempore, cum soluta nobis est eligendi optio, cumque nihil impedit, quo minus id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae.

4.2. References to figures and panels

As usual with LaTeX, it is easy to refer to a figure: see figure **??**. It is possible to refer to a specific panel in a figure, for instance figure **??** or figure **??**. Its also possible to refer to the entire figure, for instance figure **??** or figure **??**.

5. A section with table

Here is a section with a variety of tables. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates

TABLE 1. Basic table with one panel and multicolumns

	Columns 1–2		Columns 3–4	
	Column 1	Column 2	Column 3	Column 4
Line 1:	α	β	γ	δ
Line 2:	ϵ	ϕ	κ	η
Line 3:	κ	ν	π	κ
Line 4:	ψ	μ	ν	ζ

This is a note for the table. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Pellentesque nec justo aliquet, commodo nulla sed, fringilla odio. Nullam non hendrerit nisi. Curabitur et metus vel velit blandit pharetra. Morbi interdum metus a erat bibendum, nec hendrerit eros ultricies. Vestibulum vel arcu id nulla ultricies commodo. Suspendisse potenti.

repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat.

5.1. A subsection with a simple table

Table 1 is a simple table with one panel. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat. Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat.

Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat.

Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae.

6. A section with cross-references

As usual, it is possible to reference an equation, such as equation (2). It is also possible to reference a section, such as section 4, or a subsection, such as

TABLE 2. Bigger table with several panels

	Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7
A. A first panel with calligraphic letters							
Line 1:	$\mathcal{A}$	$\mathcal{C}$	$\mathcal{V}$	–	–	$\mathcal{K}$	$\mathcal{A}$
Line 2:	$\mathcal{X}$	$\mathcal{H}$	$\mathcal{O}$	–	–	$\mathcal{K}$	$\mathcal{A}$
B. A second panel with capital letters							
Line 3:	U	B	J	K	A	K	A
Line 4:	N	Y	T	L	T	K	A
Line 5:	G	S	Q	P	Q	K	A
C. A third panel with blackboard-bold letters							
Line 6:	$\mathbb{U}$	$\mathbb{B}$	$\mathbb{J}$	$\mathbb{K}$	$\mathbb{K}$	$\mathbb{K}$	$\mathbb{A}$
Line 7:	$\mathbb{N}$	$\mathbb{Y}$	$\mathbb{T}$	$\mathbb{L}$	$\mathbb{T}$	$\mathbb{U}$	$\mathbb{E}$
Line 8:	$\mathbb{G}$	$\mathbb{S}$	$\mathbb{Q}$	$\mathbb{P}$	$\mathbb{Q}$	$\mathbb{K}$	$\mathbb{P}$
D. A fourth panel with numbers							
Line 9:	1.0%	2.3%	4.5%	9.0%	9.8%	91.2%	0
Line 10:	∞	23	1	90	33	4.0%	0
Line 11:	–	–	1.2	4.4	0	9.0%	0

Note. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat.

Source. **?**, table 1 and **?**, table 2.

section 3.1. It is also possible to reference an appendix, such as appendix **??**, or a subsection in an appendix, such as appendix **??**.

Of course it is possible to reference figures, such as figure **??**, or tables, such as table 1. It is possible to refer to a specific panel in a figure, for instance figure **??** or figure **??**, and to the entire figure, for instance figure **??** or figure **??**.

7. Quotes

It is easy to quote material within the paper. Vivamus at commodo nisl. Integer sodales eros a metus efficitur, eu tincidunt leo ullamcorper. Sed fermentum tellus nec nisi lobortis, vel pharetra neque mollis. Cras non dolor ut ex semper lobortis. Vivamus sit amet sapien in elit sodales luctus. Donec a tortor ut elit consequat mollis ac id metus. Maecenas vehicula, turpis at lacinia interdum, dolor dolor pretium turpis, non fermentum purus magna at est. Quotes appear as follows:

> Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut

> enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

So this was a one-paragraph quote. Cras non dolor ut ex semper lobortis. Vivamus sit amet sapien in elit sodales luctus. Donec a tortor ut elit consequat mollis ac id metus. Maecenas vehicula, turpis at lacinia interdum, dolor dolor pretium turpis, non fermentum purus magna at est.